

DSS 210: Business Statistics

Instructor:	Liyuan Liu, Ph.D.
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Class Meetings	TR: 9:30 am-10:45am MV295

Office No.:	Mandeville Hall 349
Zoom Office Hours	Tuesdays: 3:30-6:30pm Wednesdays: 12:00-3pm To help me manage time and make sure everyone gets a chance to meet, please use the link below to book a slot: https://calendly.com/lliu-sju/30min

Course Description:

This course covers probability concepts as well as descriptive and inferential statistics. The emphasis is on practical skills for a business environment. Topics include probability distributions, estimation, hypothesis testing, simple and multiple linear regression. You will also become familiar with spreadsheet applications related to statistics and with statistical software.

Textbook: "Business Statistics: Communicating with Numbers" 4th edition (ISBN 9781260642575) by Jaggia/Kelly, or "Business Statistics: Communicating with Numbers" 5th edition (ISBN 9781266430824) by Jaggia/Kelly. You may purchase either the ebook or hard copy, used books is acceptable since we don't have Connect assignments.

Software: This course will utilize software including EXCEL. Here is the Excel download instruction:

You may browse to portal.office.com, sign in with your SJU username and SJU password.

- Click the Install Office button
- Click Office 365
- Click Install Office Apps
- Click RUN

You may download and install Office 365 on as many as **five** devices.

If you prefer, you may call 610.660.2920 for assistance or stop by the Technology Service Center (TSC) in Science Center 129 and we can assist you with the installation.

Key Topics: (Correspond to Chapters 1-10, 14 of book: Subject to change)

Descriptive Statistics

- Data Visualization, histograms, scatterplots, shapes of distributions
- Pivot tables
- Numerical summaries of data

Probability

- Basic Probability
- Discrete Distributions (Binomial)
- Continuous Distributions (Normal)
- Sampling Distributions
- Central Limit Theorem

Inference

- Confidence Intervals
- Hypothesis Testing
- Type I and II error
- Regression, SLR, MLR, and Residual Analysis
- Ethics

Exams and Grading Criteria:

Final course grade will be assigned based on the following weights:

- **In-Class Mini Quizzes:** Every chapter, 7 quizzes approximately every 2-3 weeks after we finish the entire chapter. **30%**
- **In-Class Practice Problems:** Submission will count as a grade no matter your score for each practice. **5%**
- **Mid-term Exam:** **25%**
- **Final Exam:** **25%**
- **Team Mini Excel Project:** **10%**
- **Class Participation / Adherence to Policies:** **5%**

Missed exam/quizzes Students are expected to be present for all assignments, examinations as listed on the syllabus or announced by the instructor. Make-up exams require valid documentation (this includes medical and other family emergencies). Valid documentation of an absence must be requested through the Dean's Office and presented to the instructor in order to reschedule an exam. ***There will be no exceptions made to this policy.*** Please do not request makeup exams or extended assignment deadlines based on family or leisure travel because your request ***will not*** be accommodated. Consult the University's Academic Calendar and plan accordingly.

Final Grade Distribution:

A	95-100 points	C+	77-<80 points
A-	90 - <95 points	C	73-<77 points
B+	87-<90 points	C-	70-<73 points
B	83-<87 points	D	65-<70 points
B-	80-<83 points	F	0-< 65 points

Note: Points round up to the nearest 0.5. (Eg: 86.4 is a B, 86.6 is a B+)

Canvas provides a “dumb”/unweighted average, which you should ignore. I am unable to delete that column from the Canvas Gradebook

Attendance, laptops and calculators:

Attendance and participation are essential for learning. Therefore, attendance is required for all in- person classes and synchronous online sessions. Unexcused absences are likely to result in course penalty; more than 4 unexcused absences may result in a grade of FA (Failure due to Absence) for the course.

If you are unable to attend class due to a non-COVID-19 related illness, communicate directly with me through email. If you have COVID-19 symptoms, test positive or are exposed to an unknown or non-

SJU individual, you must complete the COVID-19 Self-Disclosure Form. If the University places you in quarantine or isolation I will be notified. If you don't notify me through email and I do not receive a COVID-19 notification from the University, you will be marked absent.

If your physical absence is due to isolation or quarantine, it is your responsibility to contact me within 48 hours outlining both the date of your expected return to class and a plan to stay on pace with readings and assignments during your mandated physical absence.

Unless specifically for the purposes of solving problems using Excel and practice, laptops may not be used. Please bring your calculator to every class.

The use of laptops/cellphones to surf the internet during class will not be permitted. Additionally, please do not email or text during class.

Bonus:

Students who do an extra project can get five points added to their final semester average. The more details of the bonus project will be posted on Canvas after mid-term exam.

Email Response Time:

During the week, emails will be returned within 24 hours. Any emails sent on weekends will be returned by Sunday evening if not before.

Use of Canvas and University email:

Canvas / SJU email will be the preferred method of communication for class-related announcements,

events, changes, etc. It is the student's responsibility to check Canvas and his/her University email account regularly. Instructor is not responsible for using email accounts other than the student's assigned University account. When email instructor, please specify your section.

Students With Disabilities:

Reasonable academic accommodations may be provided to students who submit appropriate documentation of their disability. If students have need of assistance or questions with this issue, they are encouraged to contact the Office of Student Disability Services (SDS) at sds@sju.edu or by phone at 610.660.1774. The Office of SDS also provides an appeal/grievance procedure for complaints regarding requested or offered reasonable accommodations. More information can be found at: www.sju.edu/sds.

Class Recording:

Be advised that by taking this course you consent to some or all class sessions being recorded by the instructor (and in some cases with assistance by the SJU Office of Information Technology). Recordings are intended to focus on the instructor but the voices and/or likenesses of other individuals in the class may also be recorded. Video and/or sound recordings will be stored at a secure SJU video management platform for two years and will be used or distributed only for educational or academic purposes, e.g., to allow for remote learning for those students not physically present in class or unable to attend a synchronous online class session. If you have questions about the use of such technology or you do not consent to being recorded, you must discuss and resolve your concerns with your instructor before the end of the drop/add period.

Bias, Discrimination, & Harassment:

As a Catholic, Jesuit University, Saint Joseph's is committed to providing a workplace and educational environment, as well as other benefits, programs, and activities, that are free from acts of bias, discrimination, harassment, including acts of sexual harassment, sexual assault, dating and domestic violence, and stalking. We encourage anyone who has experienced this type of harm to seek help from the University by filing a report through the [incident reporting form](#), or by calling 610-660-1145.

For information about the University's response to reports of bias, harassment, and discrimination, including definitions of prohibited conduct, information regarding confidential reporting options on- and off-campus, and resources for support, please visit <https://www.sju.edu/offices/titleix-equity> or contact titleix@sju.edu / bias@sju.edu.

While I want you to feel comfortable coming to me for support, please know that I have some reporting requirements that are part of my job responsibilities at Saint Joseph's University. For example, I am required to report any student disclosures of bias, discrimination, and harassment. I will keep the information as private as I can, but I am required to bring it to the attention of the institution's Director Office of Title IX, Equity & Compliance.

Anonymous Student Feedback

The following link allows students to provide anonymous feedback directly to the Decision & System Sciences Department (DSS). Students who believe the situation they are experiencing warrants immediate attention can notify the Department via this anonymous survey. Students may also email the Department Chair (Dr. Marcello Balduccini) directly at mbalducc@sju.edu. Please note that this link is not meant to be used as a

faculty evaluation and does not replace the student evaluation at the end of the semester.

https://sju.co1.qualtrics.com/jfe/form/SV_4NSi1TCKdsHWF5X

Responsible Employee Reporting:

While I want you to feel comfortable coming to me with issues you may be struggling with or concerns you may be having, please be aware that I have some reporting requirements that are part of my job responsibilities at Saint Joseph's University. For example, if you inform me of an issue of sexual harassment, discrimination or sexual misconduct (Sexual Assault, Sexual Harassment, Sexual Exploitation, Domestic Violence, Dating Violence, or Stalking) I will keep the information as private as I can, but I am required to bring it to the attention of the institution's Title IX Coordinator, Mary- Elaine Perry. If you would like to speak with her directly, she can be reached at mperry01@sju.edu or 610-660- 1145.

Counseling and Psychological Services (CAPS) can provide confidential support and can assist you with determining how you would like to address this issue. CAPS is located at 504A Merion Gardens and can be reached at 610-660-1090. They also have walk-in hours M-F 11- 12:30 in LaFarge. REPP, Rape Education and Prevention Program provides a 24 hour confidential helpline (610-733-9650) as well. For more information about your confidential options at SJU, as well as resources in the community go to the support website.

Academic Honesty:

Saint Joseph's University encourages the free and open pursuit of knowledge; we consider this to be a fundamental principle and strength of a democratic people. To this end, SJU expects its students, its faculty, its administrators, and its staff to uphold the highest standards of academic integrity. The University expects all members of the University community to both honor and protect one another's individual and collective rights.

The [SJU Academic Honesty Policy](#), which is published both in the Student Handbook and in the University Catalog, will be strictly enforced. It is your responsibility to understand this policy. Use of unauthorized notes or assistance during an exam will result in at least failure of the assignment, if not failure of the course. Plagiarism will also result in at least failure of the assignment, if not failure of the course. Plagiarism occurs in any instance where one submits as one's own work statements or ideas taken from another source but not properly acknowledged as such. Any material copied or derived from another source must be acknowledged by giving a reference. For direct quotes, always use quotation marks and provide a reference. If you paraphrase something, you must provide a reference to acknowledge where you found the material. Internet sources, if used, must be cited.

Students are expected to include the following Academic Honesty Affirmation on each of their assignments: "I attest to the honesty of my academic work and affirm that it conforms to the standards of the Academic Honesty Policy of Saint Joseph's University."

AI Usage Policy for Academic Integrity:

Purpose: This policy is designed to clarify the expectations and limitations regarding the use of Artificial Intelligence (AI) tools in completing coursework for this class.

Scope:

- AI tools include, but are not limited to, machine learning models, chatbots (e.g., ChatGPT), AI content generators, and automated problem solvers.

- This policy applies to all assignments, quizzes, exams, and projects unless otherwise specified.

Guidelines:

- Disclosure: Students must disclose the use of any AI tools in completing coursework. Each submission should include a statement or footnote indicating the tools used and the extent of their involvement.
- Prohibited Uses:
 - Exams and Quizzes: AI tools **are not allowed** to answer questions on exams and quizzes.
 - Original Work: AI should not be used to generate entire essays, reports, or any other assignment meant to assess individual thought and analysis. Students are encouraged to use their own reasoning and capabilities.
- Permitted Uses:
 - Research Assistance: AI can be used for gathering background information, data analysis, and summarization to aid understanding.
 - Drafting Assistance: AI may be used to help draft outlines or to provide suggestions for improving grammar and style in early drafts. However, the final submission must significantly reflect the student's original work and understanding.
- Ethical Considerations:
 - Students must ensure that the use of AI does not infringe on the rights of others or violate any laws regarding data privacy and intellectual property.
 - Submissions generated with the help of AI must adhere to the principles of academic honesty and integrity.
- Consequences: Failure to comply with this policy will be considered a violation of academic integrity and will be subject to disciplinary actions as outlined in the student handbook.

Discussion and Updates:

- Students are encouraged to discuss the use of AI tools with the instructor if unsure about appropriate applications.
- This policy may be updated as new AI technologies and applications emerge.

Changes to Syllabus:

This syllabus is subject to change at the sole discretion of the instructor. Any changes will be announced class- wide in a timely manner and posted on the class website.

Inclusivity Statement:

I am committed to providing an atmosphere for learning that respects diversity. While working together to build this community I ask that we all:

- Recognize that our classroom community represents a rich variety of backgrounds and perspectives.
- Share our unique experiences, values, and beliefs.
- Be open to the views of others.
- Honor the uniqueness of our community.
- Appreciate the opportunity that we have to learn from each other in this community.
- Value each other's opinions and communicate in a respectful manner.
- Keep confidential discussions that the community has of a personal (or professional) nature.
- Use this opportunity together to discuss ways in which we can create an inclusive environment in this course and across the SJU community.

Food & Basic Needs

Your well-being is of primary importance to the learning environment. Facing financial challenges, such as difficulty purchasing textbooks, technology or other obstacles -- like securing food, basic needs, or transportation -- may impact your ability to succeed in your courses.

If you are facing any of these hardships, [please complete this form](#), which will be sent to the Office of Student Success. A staff member will work with you to connect you with University and/or other resources. For support with food and basic needs, you are encouraged to visit [HawkHUB](#) in person or submit an online order.

Finally, you are welcome to contact me to discuss specific ways in which I can support you. I am committed to helping you succeed.